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Anomalous low palaeomagnetic inclinations from Oligocene–Lower Miocene red beds of the south-west Tien Shan, Central Asia

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SUMMARY

We studied Oligocene to Lower Miocene red beds from the western margins of the South-West Ghissar Range (the westernmost part of the Tien Shan fold belt, Central Asia). A total of 19 sites were sampled on both limbs of a gentle syncline. After removal of an unstable component at 200° to 300 °C, the natural remanent magnetization in these red beds from 18 sites is accounted for by one characteristic component, ChRM, which persists up to 680 °C. Intermediate directions of the geomagnetic field were recorded at five sites; after tilt correction, the overall mean direction for the remaining 13 sites is $D=3°$, $I=30°$, $k=19$, $\alpha_{95}=8.9°$. Both the reversal and the fold tests are positive. Haematite is found to be the main remanence carrier in these red beds. The calculated mean declination fits well with the reference data (Besse & Courtillot 1991), and no rotation with respect to Eurasia is detected, in accord with geological data. In contrast, the mean inclination is about 30° shallower than expected. We argue that this result is not due to large-scale northward transport of the area. Mean inclinations showing a similar amount of shallowing were also reported from the Tadzhik and Fergana basins and Kirghiz Tien Shan from various rock types (Thomas *et al.* 1993, 1994), and we are of the opinion that neither inclination error nor compaction could have led to such a consistent pattern. The analysis of all reliable Early Tertiary palaeomagnetic results extracted from the Global PaleoMagnetic data base for the northern part of the Eurasian plate reveals that inclination anomalies are very common in this data set, whatever the rock type. The general tendency is a progressive increase in the anomaly amplitude from the Atlantic Ocean to Central Asia. The size and localization of this Tertiary field anomaly is comparable with the present-day non-dipole field anomaly. This suggests that there may have been a long-term standing non-dipole anomaly in the Tertiary Earth magnetic field, perhaps correlated to major features of the contemporaneous plate tectonics, for example the India–Asia collision.

Key words: APW (apparent polar wander), Earth's magnetic field, palaeomagnetism, Tertiary.

INTRODUCTION

In order to evaluate rotations and poleward displacements of displaced terranes within mobile belts, palaeomagnetists have commonly used the apparent polar wander paths (APWPs) of major plates as a reference. Time coverage, however, is not always adequate, and gaps or poorly defined segments still exist in many APWPs. In some cases, the number of individual poles may be sufficient, but all the results come from a limited part of the plate, thus allowing for the possibility that local effects affect the APWP over the entire plate. Such is the case for the Palaeogene part of the Eurasian APWP since all high-quality individual poles used for computation of the mean pole came solely from the British Igneous Province (Van der Voo 1993).

There are two options for obtaining a better data coverage. The first is to use a global data set transferred to the Eurasian coordinates using plate kinematics (e.g. Besse & Courtillot 1991). The second is to obtain a new regional reference from an area presumed to be unaffected by tectonic movements, but relatively close to the mobile zone where palaeomagnetic investigations are being undertaken.

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As we have undertaken a project directed towards the palaeomagnetic and structural study of Alpine deformation in Central Asia, we decided to use both options. Keeping in mind a possible comparison with the Eurasian and/or synthetic APWPs, we also studied a collection of samples from a nearby area where Alpine tectonic movements can almost certainly be ruled out. The palaeomagnetic results obtained from Oligocene–Lower Miocene rocks sampled in the Dekhkanabad area, at the boundary between the stable Turan plate and the mobile Tien Shan belt, are presented below (Fig. 1).

In all of the individual palaeomagnetic investigations that we have conducted on Tertiary rocks during the several years that this project has been running, we have found inclinations systematically shallower than those expected (Thomas *et al.* 1993, 1994). This finding made the comparison with the Eurasian reference more ambiguous, and reinforced our wish to use a local reference too. Furthermore, this intriguing feature has already been mentioned by several authors (Hailwood 1977; Westphal *et al.* 1986), and we decided to analyse it using the recent Global PaleoMagnetic (GPM) database (Lock & McElhinny 1991), taking into account the new data that we have collected in this study.

GEOLOGICAL SETTING AND SAMPLING

The area studied is situated at the western piedmont of the South-West Ghissar (Gissarsky) Range in the south-western part of the Tien Shan belt (Fig. 1). A gradual increase in Alpine deformation of Late Miocene to Pliocene–Quaternary age is observed from west to east across this range. Further eastwards in the Tadzhik basin, Alpine deformation continues to increase to the western margins of the Pamirs. In contrast, to the west almost undeformed Palaeogene (and older strata) gradually dive under a late Cenozoic cover from the South-West Ghissar Range.

In the area studied, Palaeogene and Lower Miocene

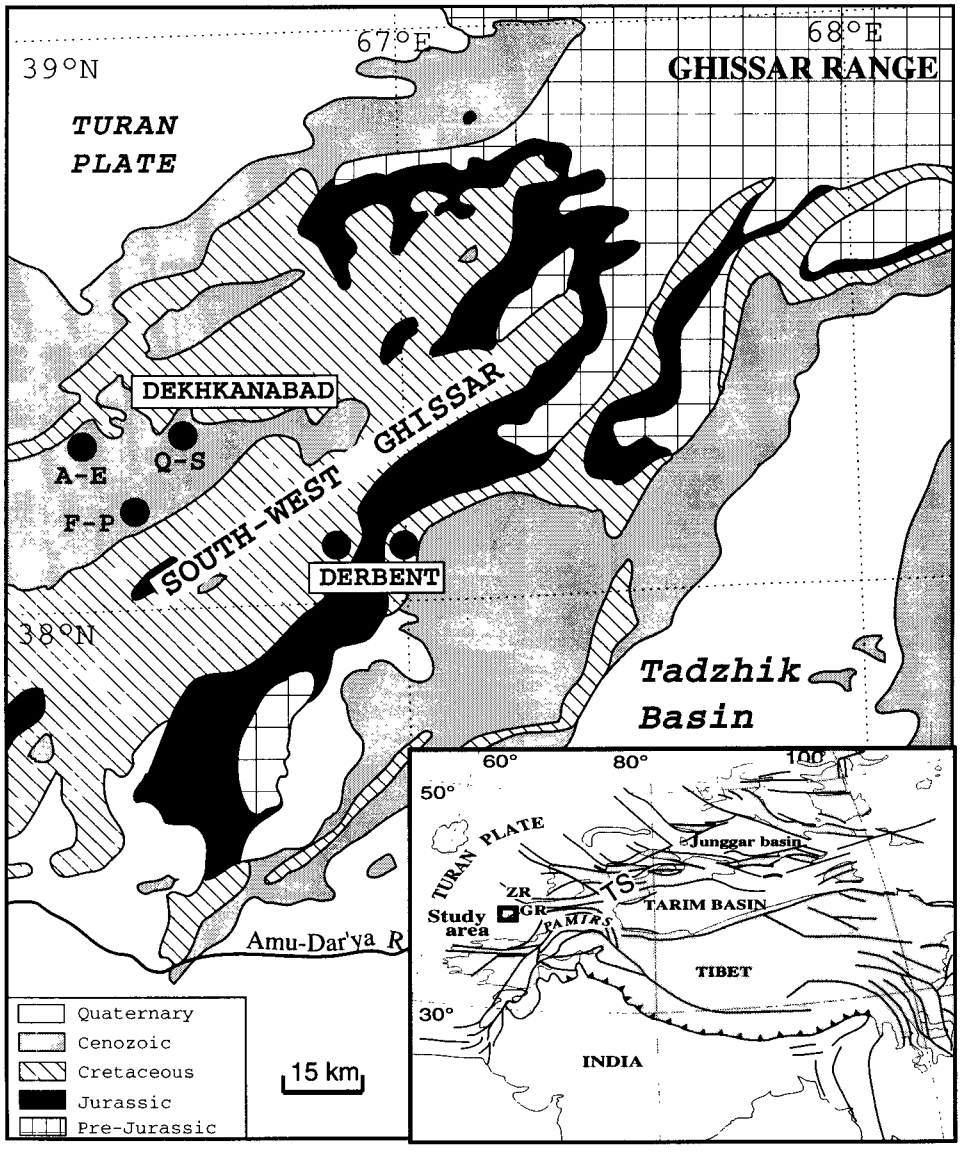


Figure 1. Geological map of the South-West Ghissar Range, with sampling localites shown as solid dots. Insert: the position of the area studied on a map of the main structural units of Central Asia. The territory with elevations above 2000 m is shaded [simplified after Cobbold & Davy (1988)]. GR: Ghissar Range; ZR: Zeravshan Range; TS: Tien Shan.

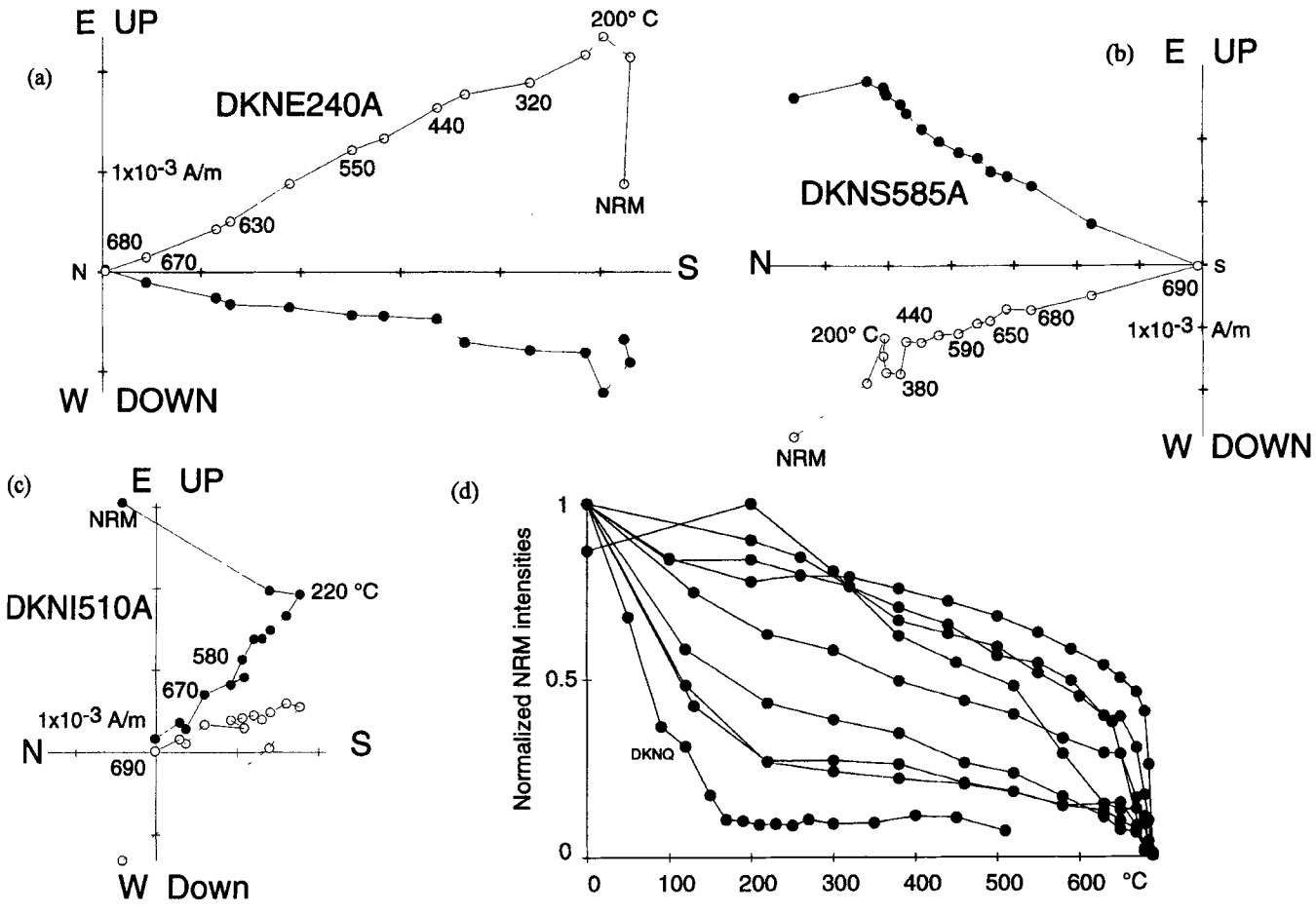


Figure 2. Representative orthogonal plots, in geographical coordinates, of thermally demagnetized samples showing characteristic components: (a) normal, (b) reversed and (c) intermediate directions. Solid (open) symbols are projections onto the horizontal (vertical) plane. Temperatures are in °C. (d) plots of NRM intensity versus temperature.

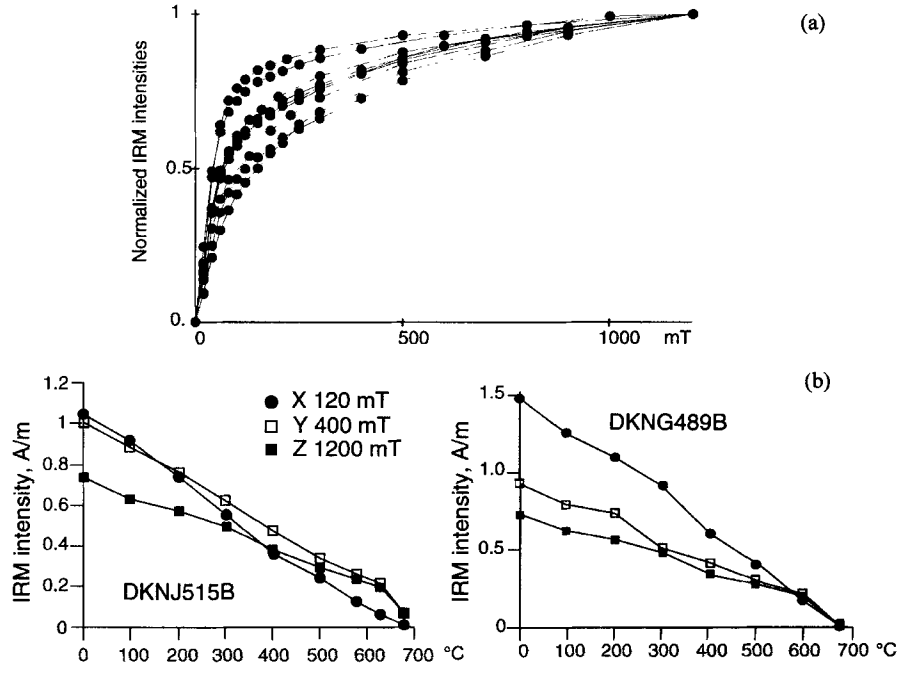


Figure 3. (a) Representative normalized IRM intensity curves; (b) IRM thermal demagnetization after Lowrie (1990) (see text).

Table 1. Palaeomagnetic results from the Dekhkanabad locality.

		IN SITU Direction				TILT CORRECTED Directions		VGP	
Site	N	D	I	k	α_{95}	D	I	LAT	Long
DKNA	8/8	189	-5	131	4.8	191	-14	-57.4	45.7
DKNB	8/8	185	-8	99	5.6	187	-17	-59.8	52.3
DKNC	7/7	182	-4	42	9.4	183	-14	-58.8	60.4
DKND	7/6	188	-17	13	19.6	186	-33	-69.1	50.0
DKNE	6/4	184	-30	67	11.3	189	-38	-71.4	39.0
DKNR	7/6	355	38	29	12.6	356	51	82.6	273.9
DKNS	8/8	28	10	197	4.0	30	21	51.6	194.2
DKNF**	7/6	324	31	20	15.3	324	19	47.0	304.5
DKNG	8/4	359	36	49	13.3	355	25	64.5	257.8
DKNH	8/7	355	60	37	10.1	348	49	77.1	299.9
DKNI**	7/5	134	-8	37	12.8	134	7	-30.4	122.8
DKNJ	7/6	5	16	59	8.8	4	4	53.6	239.6
DKNK	7/6	333	49	11	20.8	333	35	59.9	305.1
DKNL**	8/8	318	35	56	7.5	320	21	45.0	309.7
DKNM**	8/6	320	37	52	9.3	321	25	47.3	311.1
DKNN	8/4	24	44	29	17.4	16	35	66.5	205.6
DKNO	8/8	181	-52	19	13.1	176	-40	-74.2	80.0
DKNP**	8/8	358	-44	9	19.4	4	-55	16.1	243

MEANS:		IN SITU				TILT CORRECTED			
Site	N	D	I	k	α_{95}	D	I	k	α_{95}
NORTHWESTERN LIMB									
A-E,R,S	7	8	15	24	10.9	9	27	23	11.1
SOUTHEASTERN LIMB									
G-K,N,O	6	0	44	18	13.5	356	32	17	13.8
OVERALL MEAN (A-E, G-K, N, O, R, S)									
	13	5	29	14	10.6	3	30	19	8.9
FOLD TEST (McElhinny 1964)					$F_{(24,24)}$= 2.0		Ktc/Kis = 1.4		
FOLD TEST (McFadden and Jones 1981)					$F_{(2,22)}$= 3.4				
		f = 7.3					f= 1.5		
REVERSAL TEST						$F_{(2,22)}$= 3.4		f = 0.5	

**These five sites showing dual-polarity north-westerly directions were not included in the computation of the overall mean (see text).

N is the number of specimens studied/used, and the number of sites for mean directions; D is declination; I is inclination; K is the Fisher precision parameter; α_{95} is the radius of the 95 per cent confidence circle. Lat and Long are the latitudes and the longitudes of the virtual geomagnetic poles (VGPs). F is the 95 per cent critical value of F distribution for the number of degrees of freedom in brackets. Ktc/Kis is the ratio of precision parameters after/before tilt correction, f is the calculated value of F statistics.

sediments crop out on both limbs of a gentle syncline of a SW–NE strike with dip angles not exceeding 20°. It is one of the westernmost localities to be folded and uplifted by Alpine deformation, which took place here in Late Miocene and (mainly) Pliocene–Quaternary times. The oldest strata exposed are Eocene limestones that are conformably overlain by a red bed sequence of intercalated sandstones, siltstones and mudstones, usually calcareous and almost unfossiliferous; neither sequence shows any signs of penetrative deformation. The Oligocene–Early Miocene age of these red beds is established by regional correlation with other localities, where pollen and sometimes molluscs have been observed (Davidzon, Kreydenkov & Salibaev 1982).

The studied red beds are from the lower part of the sequence. In total, 19 sites from three different localities were sampled on both limbs of the fold (Fig. 1). Seven to eight cores oriented with magnetic and sun compasses were drilled at each site, which represented an apparently homogenous part of a layer up to 1 m thick and 8 m long. Both sandstones and siltstones were sampled. No continuous section could be found, and the samples were drilled from isolated small outcrops along the banks of dry creeks. The true thickness studied is about 100 m at locality 2 (sites DKNF-O) and greater for the entire sampled sequence.

During the same expedition, we also sampled Lower Cretaceous rocks in the central part of the South-West Ghissar Range (Derbent locality, Fig. 1), as well as both Lower Cretaceous and Oligocene–Lower Miocene sediments from several localities in the Tadzhik basin and the Ghissar Range. The results from these localities are presented elsewhere (Bazhenov *et al.* 1994; Thomas *et al.* 1994) and will be referred to here.

TREATMENT

One specimen of standard size per core was subjected to progressive thermal demagnetization using a Schonstedt TSD-1 demagnetizer. Samples were measured with a cryogenic magnetometer (LETI). The results were analysed with stereonets and orthogonal plots using principal component analysis. Field tests were performed at the site-mean level. With the enhanced Digico apparatus, anisotropy of magnetic susceptibil-

ity (AMS) was measured on 18 sites that yielded meaningful results.

RESULTS

Intensities of natural remanent magnetization (NRM) in these red beds range from 10^{-2} to 10^{-3} A m^{-1}. Weak-field susceptibilities range from 2.5×10^{-4} to 4.5×10^{-4} SI units. A low-temperature component of magnetization was removed after 200° to 300 °C, and, at higher temperatures, only one characteristic component, ChRM, is observed (Figs 2a–c). The only exceptions are sandstones from site DKN-Q where no stable magnetization was isolated, and this site was rejected. Blocking temperatures of the ChRMs are usually above 600 °C, thus suggesting haematite as the remanence carrier (Fig. 2d). Isothermal remanent magnetizations, IRMs, of representative samples do not reach saturation in a field of 1.2 T (Fig. 3a); quick IRM growth in the low fields, however, points to a low-coercivity phase in these rocks. Partial IRMs were then created at successively smaller fields along three orthogonal axes of a specimen and then thermally demagnetized as suggested by Lowrie (1990). The three-component IRMs were produced in fields first of 1200 mT along the Z-axis, then of 400 mT along the Y-axis and finally of 120 mT along the X-axis. Irrespective of the direct field in which the IRM was created, the IRM is completely destroyed at well above 580 °C for the three curves (Fig. 3b). These observations suggest that haematite of variable coercivity, presumably because of very variable grain sizes, is the main carrier of the remanence in these rocks and that some amount of magnetite may also be present (sample DKNJ515B, Fig. 3b).

Both normal and reversed directions of the ChRM are observed (Table 1); intermediate directions are also present at one locality. The latitudes of virtual geomagnetic poles (VGP) for five sites (DKN-F, I, L, M, P) are lower than, or very close to, 45° (Fig. 4), although values of greater than 60° are expected. Results from these sites were therefore interpreted as transitional-field values and excluded from the calculation of the overall mean direction.

The presence of both polarities and intermediate directions implies that the time delay of the acquisition of magnetization in those rocks is rather short. In order to check if the dispersion of the remaining 13 site-mean VGPs indicates a sufficient time coverage, the angular standard deviation of the field, SF, from between-site dispersion was calculated and compared with the expected angular dispersion from recent models of secular variation for the time period 22.5–45 Ma (McFadden *et al.* 1991). The VGP population yields the dispersion $SF = 15.2°$ with 95 per cent confidence limits of 21.1 and 11.9 (Cox 1969) (with average site dispersion $Sw = 13.1°$, average number of samples per site $n = 6.3$, total angular dispersion of VGPs, $ST = 16$ and $SF^2 = ST^2 - Sw^2/n$). For the present-day latitude of the study area, the model of secular variation (McFadden *et al.* 1991) predicts an angular dispersion of the field about 18.9°, which is close to the observed value and falls within the statistical confidence limits. Although the number of sites is not very large, this result seems to indicate that the time interval covered by our sampling provides an adequate averaging of geomagnetic secular variation.

In stratigraphic coordinates, the overall mean direction for the 13 site means is D = 3°, I = 30°, $\alpha_{95} = 8.9°$, corresponding to a palaeomagnetic pole at Lat = 67 °N, Long = 239 °E, dp = 5.5°, dm = 9.9° (where dp and dm are parameters of the ellipse of confidence associated with the pole position). The concentration parameter for this data set is rather low, both *in situ* and after tilt correction, whereas bedding attitudes of the studied beds are very gentle. For these reasons, the fold test after McElhinny (1964) is inconclusive. In contrast, the fold test after McFadden & Jones (1981) is positive, pointing to a pre-folding age of magnetization at the 95 per cent confidence level (see Table 1 for the statistical parameters), as is the

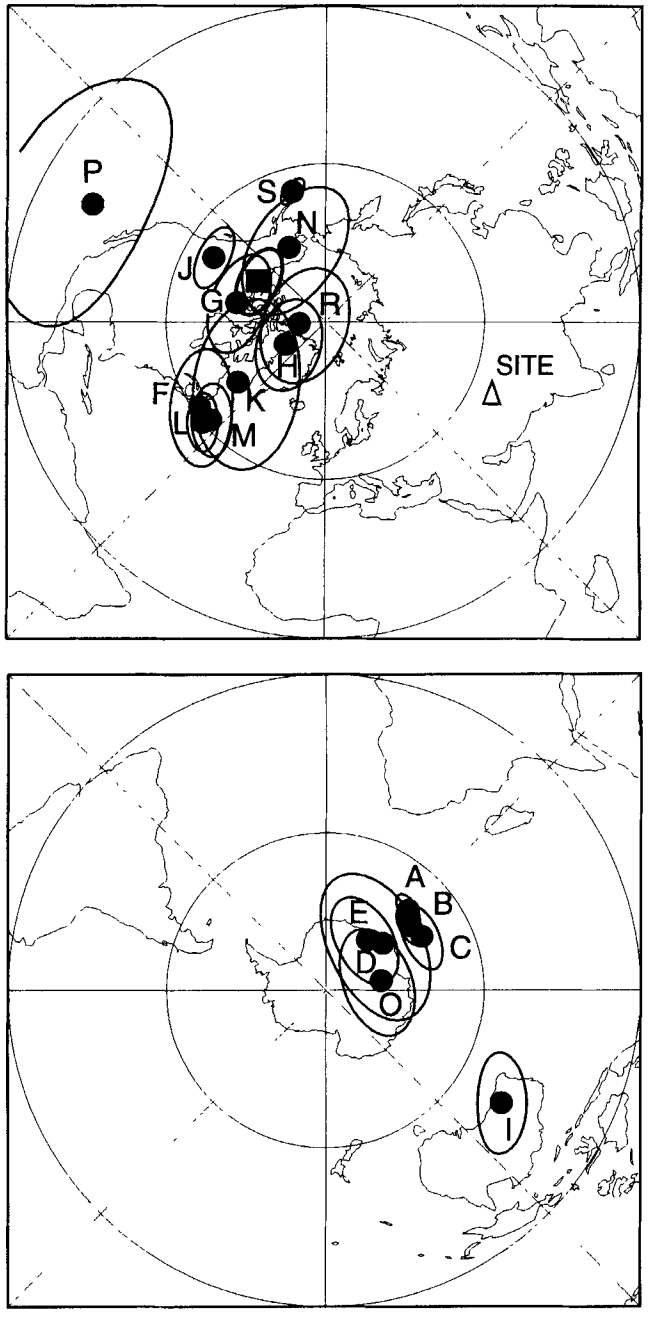


Figure 4. Plot of site-mean VGPs with associated ovals of confidence, as listed in Table 1, calculated using site-mean directions in stratigraphic coordinates. The black square denotes the mean VGP (238.7°E, 68.3°N); the white triangle denotes the area studied.

reversal test on the tilt-corrected site means. Thus we conclude that the Oligocene–Lower Miocene red beds retained an NRM component whose age is likely to be close to the age of the rock.

The anisotropy of magnetic susceptibility, AMS, was measured on 75 specimens. The mean magnetic tensors calculated for the sampled localities (first locality: sites DKNA to E; second: sites DKNF to P; third: sites DKNQ to S) have an oblate shape, i.e. maximum (*K*max) and intermediate (*K*int) anisotropy coefficients are almost equal. So, if some preferred orientations of *K*max and *K*int are noticeable (Fig. 5a), these are hardly significant. This suggests that almost no preferred reorientation of magnetization carriers in the horizontal plane has taken place since the NRM acquisition. The average magnetic foliations are almost parallel to the bedding (Fig. 5a), indicating a depositional magnetic fabric (Hrouda 1982). The foliation for the three localities varies from low values of 3 per cent at localities 1 and 2 to moderate ones of 6 per cent at locality 3. At the site level, no correlation is observed between the inclination shallowing and anisotropy parameters (see Figs 5b and c) or sample lithologies.

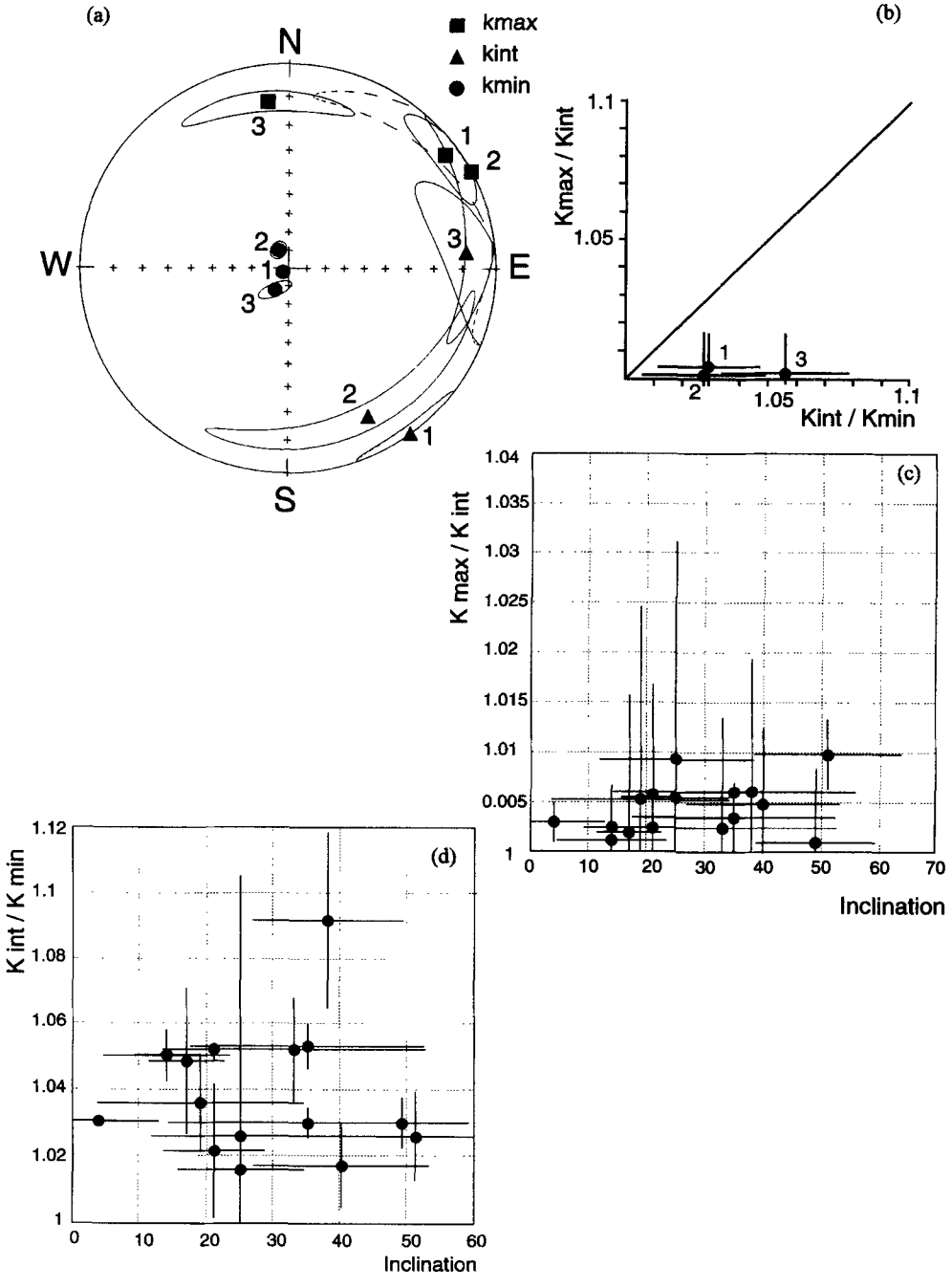


Figure 5. Stereographic projection of the principal axes of mean AMS tensors (tilt corrected) for the three localities sampled. Locality 1: sites A to E; locality 2: sites F to P; locality 3: sites Q to S. (b) Cross-plot of lineation versus foliation parameters; (c) and (d) cross-plot of lineation (foliation) parameters versus observed inclination (*I*) for all sites except Q, I and P.

INTERPRETATION OF THE CHARACTERISTIC MAGNETIZATION

Because the age of the studied rocks is between 30 Ma and 20 Ma, we calculated a reference direction for our locality (*ca.* 38.5 °N, 66.5 °E) as the mean between the 30 Ma and 20 Ma Eurasian mean poles (Besse & Courtillot 1991). This direction is $D = 10.4°$, $I = 59.6°$, $\alpha_{95} = 2.3°$; the reference direction derived from the Eurasia-only APWP of Van der Voo (1993) is similar within the error limits.

The mean declination of our data is not statistically different from that of the reference, the parameter R (Demarest 1983) being equal to $7 \pm 9°$. (Fig. 6a). It implies that no significant rotation of the studied area took place during Alpine deformation, in accord with geological data. This conclusion is also in agreement with the results from Lower Cretaceous red beds from Derbent, even further to the east (Bazhenov *et al.* 1994), where the R value of $5 \pm 5°$ with respect to the APWP of Besse & Courtillot (1991) is within the error limits of the method. Taken together, these data justify the choice of the Dekhkanabad locality as a regional reference.

In contrast, the mean inclination is much shallower (Fig. 6b), giving a flattening value F of $29 \pm 8°$. We have also recovered similar low inclinations, giving flattening values F from $24 \pm 11°$ to $30 \pm 10°$, on average 27 °, from Oligocene to Lower Miocene rocks from six localities spread over 250 km in the Tadzhik basin (Thomas *et al.* 1994). Moreover, we have reported similar F values (21 ° on average) from both sediments and a few basalts of Eocene to Early Miocene age from the Fergana basin and North Tien Shan (Thomas *et al.* 1993).

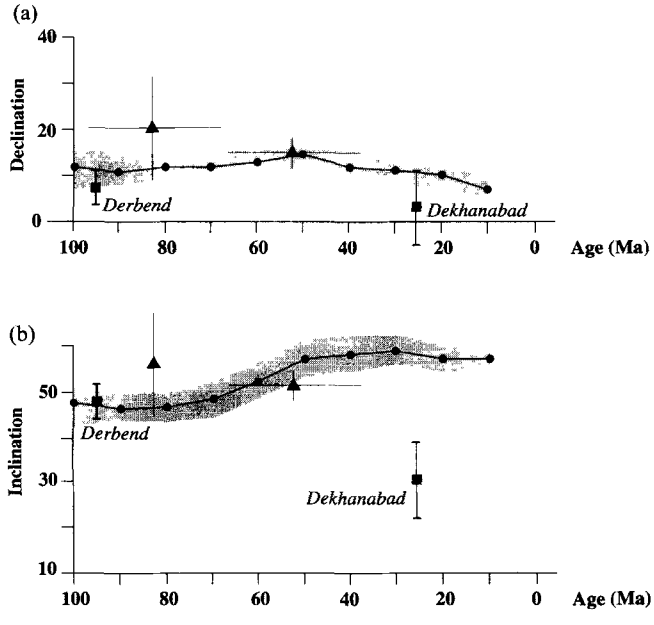


Figure 6. Comparison of our palaeomagnetic data (filled square) obtained near the Turan platform with apparent polar wander paths (APWPs). Graphs show palaeomagnetic declination (a) and inclination (b) as a function of age. The APWP of Courtillot & Besse (1990) is represented by the segmented curve joining mean values (filled circles) between 95 per cent confidence limits (shaded strips). The APWP of Van der Voo (1990) is represented by two filled triangles with error bars. Paths were calculated for a point at 38°N, 67°E. Derbent: Upper Cretaceous formations (Bazhenov et al. 1994); Derkanabad: Oligocene to Lower Miocene formations.

Thus this inclination anomaly appears to be a regional phenomenon.

Converted to latitude, the Dekhkanabad mean inclination would imply a northward shift of about 2 500 km of the studied locality since the Early Miocene. However, there is no geological support for such a large movement. Indeed, the studied area is situated on the westernmost margins of the Tien Shan highlands. Only the western part of the Zeravshan Range protrudes more to the west into the Turan plate (ZR, Fig. 1), and it is the only zone to the north of Dekhkanabad where any Alpine deformation may have been accommodated. However, the width of this range, just a few tens of kilometres, and the moderate to low magnitudes of Late Cenozoic folds and thrusts here can account for only a marginal northward movement. In general, the Cenozoic shortening of the entire Tien Shan is thought not to exceed 200 km (Avouac & Tapponnier 1993; Avouac *et al.* 1993).

The Turan plate itself is not tectonically dead, as indicated by a series of strong Gazli earthquakes several hundred kilometres to the WNW of Dekhkanabad. The Cenozoic cover over this plate, however, is either completely undeformed or locally affected by gentle folding and low-magnitude thrusting (Khain 1984), thus not allowing the accommodation of considerable horizontal movements. A westward extrusion of the studied area as a consequence of the India–Eurasia collision (Molnar & Tapponnier 1975) could also have resulted in a negligible inclination anomaly. In general, there are no traces of geological structures that are compatible with the inferred poleward movement of 2 500 km. Moreover, palaeomagnetic data from older rocks from the same areas reveal no inclination anomaly. Cretaceous inclinations from both the South-West Ghissar Range (Derbent locality, Fig. 1) and the Tadzhik basin (Pozzi & Feinberg 1991; Bazhenov *et al.* 1994) agree very well with the reference directions (Besse & Courtillot 1991).

ANALYSIS OF THE SHALLOW INCLINATION DATA SET

Both geological and palaeomagnetic data imply that shallow inclinations in Oligocene–Lower Miocene rocks cannot be accounted for by poleward movements. The regional character of the inclination anomaly and the internal consistency of our palaeomagnetic data of similar age also preclude any possibility of primary tilts of the studied strata or of a bias due to an unidentified post-folding component or internal deformation. In order to assess better the geographical extent and time interval of the occurrence of these shallow inclinations, we have conducted a systematic compilation of all palaeomagnetic data published for the same period of time, for the stable part of the Eurasian plate. This analysis was made with the aid of the Global PaleoMagnetic data base (Lock & McElhinny 1991). We used the most recent distribution, which includes the Russian data and all palaeomagnetic results published up until 1993. From the full data base, we have extracted all entries satisfying the following criteria:

(1) the magnetization age is well constrained in the interval 20–50 Ma;

(2) the site does not belong to a known largely displaced terrane (e.g. India);

(3) there has been representative sampling;

(4) the demagcode ≥ 2 (*cf.* Lock & McElhinny 1991 for the significance of the demagcode);

(5) the results show statistical significance (e.g. $k \geq 10$ or $\alpha_{95} \leq 15°$).

The palaeomagnetic results passing these criteria are listed in Table 2 for the European part and Table 3 for the Asian part, and are identified by their GPM result number. For more information on each entry, please refer to the GPM data base using this key number. For comparison with the reference APWP, the palaeomagnetic results were assembled in three age groups (20, 30 and 45 Ma). Then, we calculated expected inclinations and palaeolatitudes, using the reference APWP of Besse & Courtillot (1991), and the differences between expected and observed inclinations and palaeolatitudes. For each group, a mean difference and its standard deviation were also computed. A significant shallowing of inclination is detected when this mean difference is greater than its standard deviation. This is the case for the 30 Ma group of the European data set, and for all three groups of the Asian data set. As indicated in the tables, the rock types for these data include both volcanic (V) and sedimentary (S) rocks.

To create a better image of the distribution of the shallow inclinations, we obtained a graphical representation (Fig. 7) using the GMT software of Wessel & Smith (1991). First, we computed the mean inclination shallowing for a 5° grid. Then, we interpolated the results using the 'Surface' tool (Smith & Wessel 1990) over the entire area under analysis and imaged the resulting map with a grey scale from minimal to maximal inclination shallowing. To help visualization, we made contour plots for every 10°. The data used were the GPM results listed in Tables 2 and 3, plus our own Asian data set, also listed in Table 3. Each site used is indicated with a small dot on Fig. 7.

The most striking feature of this figure is the progressive

Table 2. Lower Tertiary palaeomagnetic results from Europe for three different age groups.

GPM	Site	Site	Age	Rock	Observed		Expected		ΔInc	ΔLat
resultno	Lat	Long	(Ma)	Type	Incl	Paleolat	Incl	PaleoLat		
582	35,0	33,0	20	S	58,0	38,7	50,8	31,5	-7,2	-7,2
1248	39,5	26,5	20	V	51,0	31,7	54,7	35,2	3,7	3,5
2686	39,5	8,7	20	V	55,3	35,8	52,9	33,5	-2,4	-2,3
2761	49,7	12,1	20	V	51,4	32,1	62,5	43,9	11,1	11,8
2793	40,5	8,7	20	V	43,5	25,4	54,0	34,5	10,5	9,1
2858	39,1	8,4	20	V	25,3	13,3	52,5	33,1	27,2	19,8
2960	40,9	8,8	20	V	50,5	31,2	54,4	34,9	3,9	3,7
3167	49,5	20,5	20	I	73,2	58,9	63,0	44,5	-10,2	-14,4
3304	51,0	14,7	20	V	63,2	44,7	63,8	45,5	0,6	0,8
3389	48,0	21,0	20	V	62,3	43,6	61,9	43,1	-0,4	-0,5
3603	50,5	7,5	20	V	62,0	43,2	63,0	44,4	1,0	1,2
4121	45,0	36,0	20	S	48,0	29,0	60,7	41,7	12,7	12,7
6272	42,6	9,0	20	S	63,3	44,8	56,0	36,6	-7,3	-8,2
Mean									**3,3**	**2,3**
Std dev									**10,1**	**9,4**
419	43,4	12,6	30	S	45,0	26,6	57,8	38,4	12,8	11,8
537	39,8	20,9	30	S	38,8	21,9	55,5	36,0	16,7	14,1
667	40,7	8,5	30	V	51,0	31,7	54,8	35,3	3,8	3,6
749	40,0	8,5	30	V	52,4	33,0	54,1	34,6	1,7	1,6
781	50,3	7,0	30	V	59,1	39,9	63,0	44,5	3,9	4,6
1555	40,2	23,8	30	I	31,0	16,7	56,2	36,8	25,2	20,1
2287	50,1	11,4	30	V	59,0	39,8	63,4	44,9	4,4	5,1
2292	45,5	11,7	30	V	52,5	33,1	59,6	40,4	7,1	7,3
3117	51,0	16,3	30	V	61,7	42,9	64,5	46,4	2,8	3,5
3173	40,5	8,5	30	V	42,0	24,2	54,6	35,1	12,6	10,9
4166	45,0	40,0	30	S	53,0	33,6	62,5	43,9	9,5	10,3
6239	40,0	20,2	30	L	27,8	14,8	55,6	36,1	27,8	21,3
Mean									**10,7**	**9,5**
Std dev									**8,7**	**6,5**
418	43,4	12,6	45	S	41,0	23,5	54,8	35,3	13,8	11,8
723	37,1	14,5	45	S	35,5	19,6	48,3	29,3	12,8	9,7
884	41,7	2,5	45	S	51,8	32,4	51,9	32,5	0,1	0,1
1235	41,0	26,0	45	V	46,5	27,8	54,3	34,8	7,8	7,0
1358	49,2	22,2	45	S	40,1	22,8	61,2	42,3	21,1	19,5
1448	39,5	20,8	45	L	42,5	24,6	51,9	32,5	9,4	7,9
1550	40,0	18,5	45	S	50,6	31,3	52,1	32,7	1,5	1,4
2000	56,5	-5,8	45	I	61,4	42,5	64,6	46,5	3,2	4,0
2239	51,2	-4,7	45	I	62,7	44,1	60,4	41,4	-2,3	-2,7
2293	45,5	11,7	45	V	36,7	20,4	56,7	37,3	20,0	16,9
2846	48,0	9,0	45	V	61,5	42,6	58,8	39,5	-2,7	-3,1
3456	55,0	-4,0	45	I	59,3	40,1	63,6	45,2	4,3	5,1
4172	43,0	46,0	45	S	52,0	32,6	59,2	40,0	7,2	7,4
5711	41,5	36,4	45	S	34,4	18,9	56,3	36,9	21,9	18,0
6113	40,8	30,9	45	V	55,8	36,3	54,8	35,3	-1,0	-1,0
6238	40,0	20,2	45	L	39,0	22,0	52,4	33,0	13,4	11,0
6273	42,6	9,0	45	S	52,5	33,1	53,6	34,1	1,1	1,0
7014	49,4	19,2	45	S	58,8	39,5	61,0	42,1	2,2	2,6
Mean									**7,4**	**6,5**
Std dev									**8,1**	**7,0**

Legend. The entries are numbered as in the Global PaleoMagnetic data base (Lock & McElhinny 1991); S, V, I and L are for sandstones, volcanics, intrusives and limestones, respectively. ΔInc and ΔLat are calculated as the difference between the expected and observed values.

Table 3. Lower Tertiary palaeomagnetic results from Asia for three different age groups.

GPM	Site	Site	Age	Rock	Observed		Expected		ΔInc	ΔLat
resultno	Lat	Long	(Ma)	Type	Incl	Paleolat	Incl	PaleoLat		
783	33,3	36,3	20	V	34,2	18,8	49,3	30,2	15,1	11,4
2148	38,5	48,0	20	V	35,5	19,6	56,2	36,8	20,7	17,2
3980	60,0	83,0	20	S	61,0	42,1	75,4	62,5	14,4	20,4
4080	40,5	70,7	20	S	41,0	23,5	60,8	41,8	19,8	18,3
Mean									**17,5**	**16,8**
Std dev									**3,2**	**3,9**
793	18,0	43,2	30	V	15,2	7,7	32,7	17,8	17,5	10,1
1193	22,0	40,0	30	V	26,4	13,9	37,9	21,3	11,5	7,4
1497	32,2	52,2	30	S	34,2	18,8	52,6	33,2	18,4	14,4
3226	36,8	30,3	30	S	37,0	20,6	53,9	34,4	16,9	13,8
4170	38,4	70,4	30	S	32,0	17,4	61,0	42,1	29,0	24,7
Mean									**18,7**	**14,1**
Std dev									**6,4**	**6,6**
1194	22,0	40,0	45	V	3,2	1,6	33,6	18,4	30,4	16,8
3330	39,0	39,5	45	SV	48,0	29,0	54,5	35	6,5	6,0
4146	40,7	44,4	45	V	51,0	31,7	56,9	37,5	5,9	5,8
4171	39,4	73,3	45	S	39,0	22,0	60,6	41,6	21,6	19,6
6837	41,0	44,6	45	V	55,4	35,9	57,2	37,8	1,8	1,9
Mean									**13,2**	**10,0**
Std dev									**12,2**	**7,7**
					Our Studies					
Issyk Kul	42,3	78,2	30	VS	49,0	29,9	65,0	47	16,0	17,1
Naryn	41,2	75,0	30	S	37,0	20,6	63,8	45,4	26,8	24,8
Fergana	41,0	71,0	30	S	34,0	18,6	63,2	44,7	29,2	26,1
Derkhanabad	38,6	66,4	20	S	30,0	16,1	58,6	39,3	28,6	23,2
Tadjik Basin	38,0	69,0	30	S	33,0	18,0	60,5	41,5	27,5	23,5
Mean									**25,6**	**22,9**
Std dev									**5,5**	**3,5**

The same notation as in Table 2.

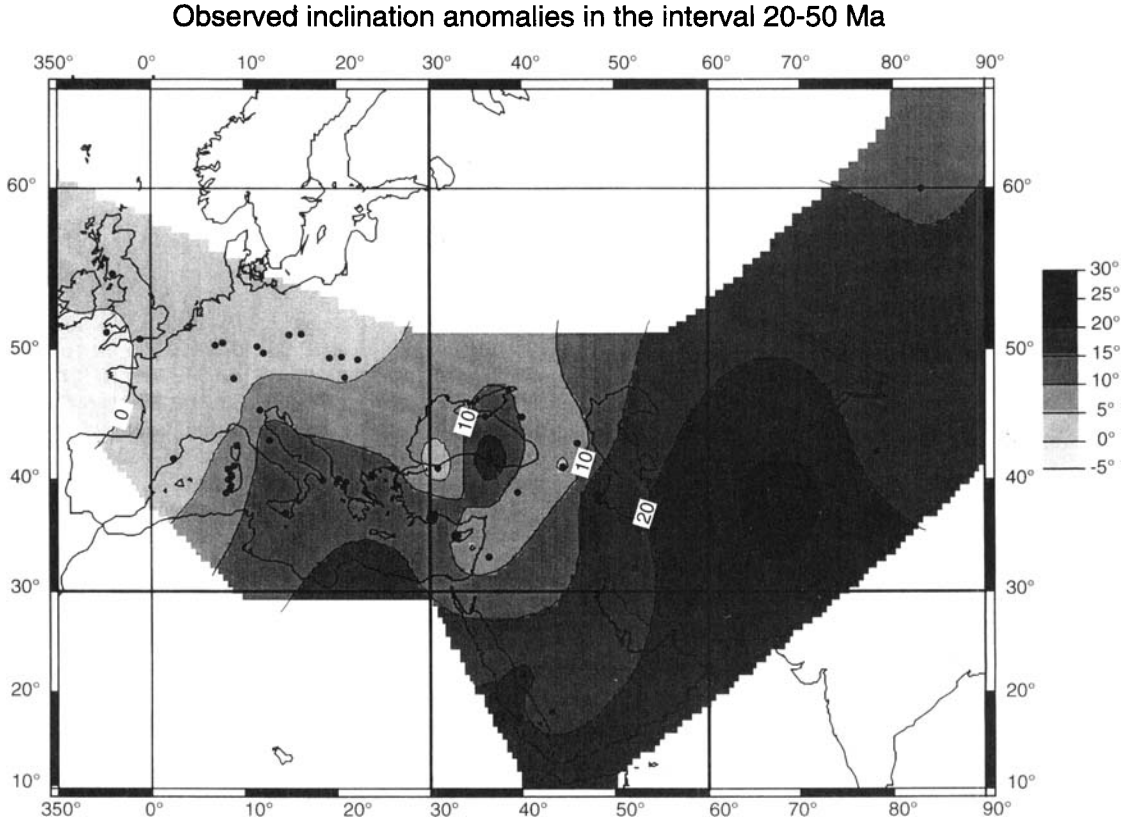


Figure 7. Contour plot of the inclination anomalies for the data set extracted from the GPM data base. Black dots represent sampled localities.

increase in the inclination anomaly, from near 0° on the Atlantic margin, to about 10° in Central Europe and reaching a maximum of 25° in Central Asia. The geographical extent of this anomaly is therefore quite comparable to the present-day non-dipole field (NDF) anomalies, and even at present there is a large NDF anomaly centred in Siberia and extending westwards up to the Atlantic Ocean. The amplitude of present-day NDF anomalies can reach about one-half of the dipole field intensity (as for example, over the South Atlantic Ocean), which can produce a local inclination anomaly of about 20–25°. The common features between the present-day NDF and our palaeomagnetic inclination anomaly suggest that there may have been long-lived NDF components in the Tertiary Earth magnetic field. If this is true, Fig. 7 would be one of the first maps of components of an ancient field of an order higher than the axial centred dipole field. In other words, with the accumulation of data, the palaeomagnetic technique would begin to be able to recover field structures of smaller scale. A further consequence would be that, since the shallow inclinations are observed with both polarities, the non-dipole components that would produce them should be reversed themselves, together with the main dipole field. Finally, we could try to relate the persistance of a NDF anomaly in this area to the most prominent plate-tectonic feature, i.e. the plate collision between India and Asia and associated subduction, through coupling mechanisms that are still to be elucidated.

CONCLUSION

Processes usually invoked to account for low inclinations are magnetization acquisition and/or compaction. Pronounced inclination error without deviation of mean declination has indeed been observed in modern red deposits (Tauxe & Kent 1984). However, similar F values (higher than 20°) were found in red beds (mostly sandstones and some siltstones) and a few basalt flows at localities more than 1000 km apart (Thomas *et al.* 1993, 1994). The ChRM in these red beds is almost everywhere carried by haematite, which may well be of chemical origin, while inclination error is possible only for grains of detrital origin. On the other hand, the AMS data (Fig. 5a) favour some compaction in the studied rocks. But, again, the observed anisotropy is too weak to account for the entire inclination anomaly; also, low inclinations were recovered from sandstones, where compaction is negligible. We think that these two mechanisms might have led to some inclination shallowing, but hardly as uniform, strong and omnipresent as in our case.

As a conclusion we would like to point out that the inclination anomaly in Palaeogene rocks has been recognized over a much larger area than just the Tien Shan and adjacent basins. This anomaly was first described by Hailwood (1977) as persisting for at least 53 Ma of the time-averaged departure of the geomagnetic field from the centred axial configuration, identified for the last 25 Ma by Wilson (1970). It has also been found over a part of the Alpine belt west of the Pamirs (Westphal *et al.* 1986). Since then, new results from the Eastern Mediterranean (Kissel *et al.* 1987, 1993), North Tien Shan (Thomas *et al.* 1993, 1994) and South China (Huang & Opdyke 1992) have extended its geographical limits and confirmed its existence. For an individual result or several results from a limited area, it is possible to attribute this anomaly to peculiarities of remanence acquisition in sediments or tectonic movements, for instance, as has already been done for the Aegean region (Beck & Schermer 1994); in each case, however, an explanation can be applied only to a limited number of results. We think that the inclination anomaly should have the same origin everywhere with, of course, local contributions from other processes. The only attempt to make a uniform interpretation was by Westphal (1993), who suggested that the anomaly may be explained by co-existing and mutually oscillating axial and inclined dipoles. This model, however, contradicts many other facts, for instance the global consistency of mean Palaeogene poles from major plates. We propose here that long-lived non-dipole components of the geomagnetic field, which are known to exist (Merril & McElhinny 1983), can be another possible explanation. Using axial quadrupole and octupole components of about 10 per cent to 15 per cent relative intensity, differences of 10° to 14° in inclination, compared with a pure axial dipole, can be produced for sites at a latitude of around 35°N. However, such amplitudes for the non-dipole components are larger than those already proposed for the Tertiary (Schneider & Kent 1990; Coupland & Van der Voo 1980). Finally, we would like to emphasize that the long-lived NDF components may have relationships with plate tectonic processes that only research oriented towards the understanding of coupling mechanisms between the lithosphere, the asthenosphere and the core–mantle boundary could attempt to resolve.

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